Lesion-Aware Domain-Specific Augmentation as an Alternative to Generic Mix-Based Methods for Tomato Leaf Disease Detection

THE UNIVERSITY OF SOUTHERN MISSISSIPPI

Md Tanjimuzzaman Turan and Partha Sengupta
School of Computing Sciences and Computer Engineering

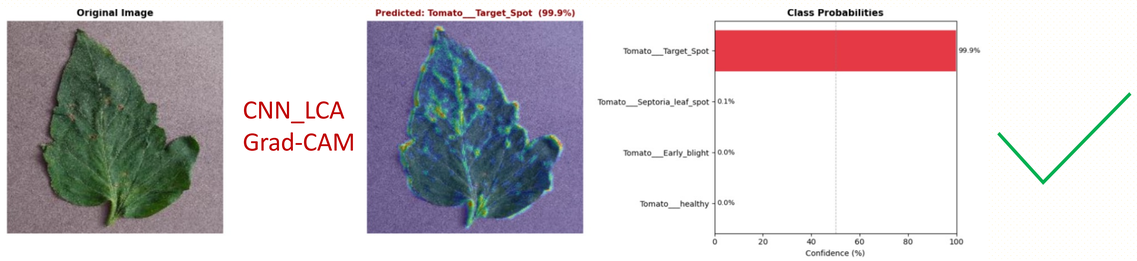
INTRODUCTION

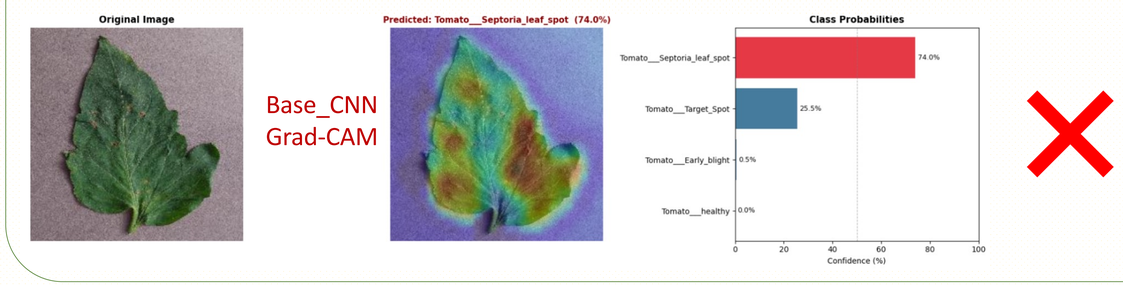
Plant disease models are usually trained on full leaf images, so they can "cheat" by using background color and global leaf appearance instead of truly learning disease lesions, especially small or early-stage spots. Popular domain-agnostic augmentations like Mixup and CutMix can improve robustness, but they also create unrealistic leaf–lesion blends that may distort these subtle cues. To address this, we generate simple lesion cue maps (from color and local-contrast rules, without manual labels) and apply a lesion-aware, domain-specific augmentation that dims non-lesion regions while preserving lesion geometry, testing whether this improves tomato leaf disease classification over a standard CNN baseline.

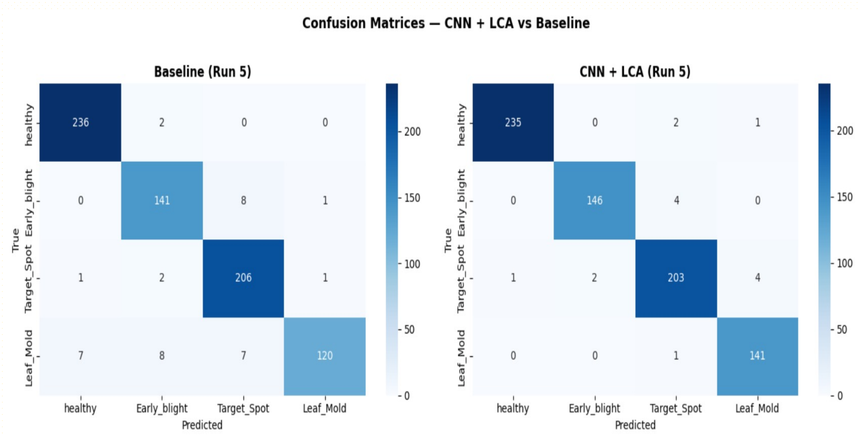
Data & Method

The PlantVillage tomato leaf dataset (healthy, Early blight, Target Spot, Leaf Mold) was used, with images resized to 256×256 and split into disjoint training, validation, and test sets. Lesion cue maps were generated using simple color and local-contrast rules, then a lesion-aware augmentation was applied by dimming non-lesion pixels while preserving lesion geometry. A standard CNN was trained with and without this augmentation under identical settings, and validation/test accuracy and per-class F1 scores were compared across multiple runs. Grad-CAM visualizations were additionally inspected to assess whether the lesion-aware model concentrated more on true disease regions than the baseline.

Using a lesion-aware augmentation that dims non-lesion pixels, CNN learns to focus on true disease lesions and achieves higher test accuracy on tomato leaf diseases compared to baseline CNN Model

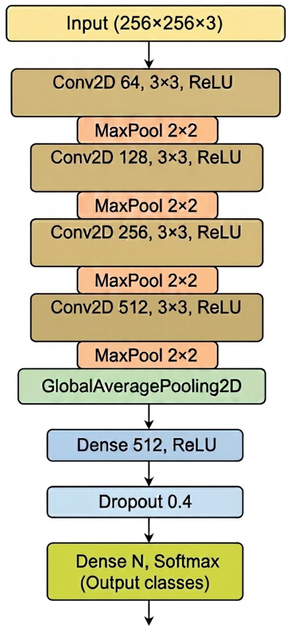






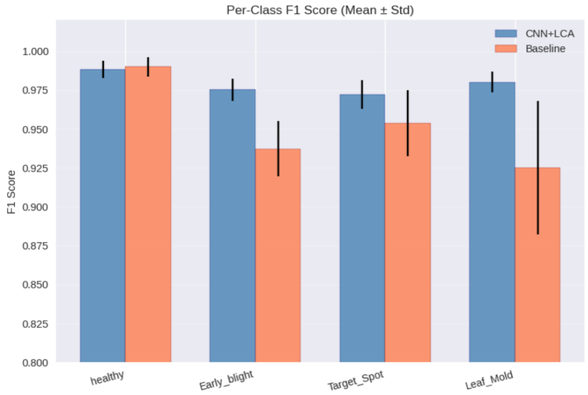
Both models perform well overall, but the Lesion-Aware CNN correctly recovers several misclassified Early blight and Leaf Mold samples, suggesting improved sensitivity to subtle lesion patterns.

CNN Architecture



RESULT

Proposed lesion-aware augmentation consistently improved performance over the plain CNN baseline. Across multiple training runs, the lesion-aware model achieved higher validation and test accuracy and better per-class F1 scores, especially for diseased classes. Confusion matrices show fewer misclassifications between similar tomato diseases when lesion cues are used.



Conclusion

The lesion-aware augmentation improves tomato leaf disease classification without changing the underlying CNN architecture, indicating that simple domain-specific cues can meaningfully help the model focus on real lesions. These findings support using lesion-focused, plant-aware augmentations rather than purely generic image transforms and motivate future work on attention-guided or multi-task lesion-aware models.

References

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